David Alejandro Torres Torres

Mechatronics Engineer | Swarm Robotics & Multi-Robot Systems Researcher

CONTACT

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SKILLS

Research Areas

- Swarm robotics & collective intelligence
- Multi-robot systems
- Autonomous navigation
- Robotic manipulation

Robotics Stack

- ROS 1/2, Gazebo, ARGoS
- URDF/Xacro, TF2, RViz, Nav2
- Simulation-to-control pipelines

Programming & Control

- C++, Python, MATLAB/Simulink
- State machines, distributed & closed-loop control
- Embedded systems (ESP32, Arduino)
- RTOS, real-time device control

Industrial Automation

- PLC-based control (Siemens TIA Portal)
- HMI integration, VFDs, sensor pipelines

Mechatronics & Design

- Actuator & mechanism design
- PCB design (EasyEDA, Proteus)
- CAD: SolidWorks, Fusion 360, Inventor
- Cyber-physical system integration

PROJECTS

Robotic Surface Printing System (2024).
Modular surface-printing attachment for a TurtleBot3 mobile robot for automated printing on planar surfaces.

Quadruped Leg Controller Platform (2023).
Dynamic test platform for quadruped robot legs using impedance-based control, validated in MATLAB Simscape Multibody.

ABOUT

Mechatronics engineer with hands-on research experience in swarm robotics, multi-robot systems, and decentralized control. My work spans the full mechatronic development cycle—from mechanical and electronic design to embedded software and multi-fidelity simulation—with emphasis on collective robotic systems for construction, inspection, and monitoring applications.

PROFESSIONAL EXPERIENCE

Research Assistant — Swarm Robotics & Collective Behavior 2024 – Present (Part-time Research)
Collective Robotics and Artificial intelligence Laboratory (CoRAL), ESPOL

- Conducted research on decentralized multi-robot systems under Dr. Christian Tutiven (ESPOL) in collaboration with Prof. David Garzón Ramos (UCD, Ireland), focusing on swarm robotics and collective construction.
- Designed and implemented cooperative robot behaviors, task allocation mechanisms, and distributed task execution pipelines using ROS-based platforms.
- Developed multi-fidelity simulation infrastructures combining ARGoS for large-scale swarm experiments and ROS/Gazebo for physically grounded interaction.
- Contributed to the full mechatronic development of the Qupa swarm robot: mechanical design, custom PCB electronics, embedded software (ESP32), and control systems.
- Built the foundational swarm robotics simulation platform at ESPOL (ROS 1 simulators, URDF models, Gazebo plugins, C++ swarm behaviors), which evolved into the CoRAL research infrastructure.

Research Assistant — Autonomous Robotics 2023 – 2024 (Remote)
Robotics and Intelligent Systems Engineering (RISE) Lab, Sungkyunkwan University (SKKU), South Korea

- Developed autonomous navigation pipelines for mobile robots using ROS 2 and Gazebo under the supervision of Prof. Nabih Pico.
- Designed Nav2-based planning and control architectures, and built simulation environments for benchmarking navigation controllers.
- Conducted comparative evaluations of learning-based and traditional control strategies based on performance, stability, and response behavior.

Automation Engineer — Industrial Systems 2023 – 2025 (Hybrid, Part-time)
WG Selcon, Ecuador

- Designed and deployed PLC-based industrial automation systems (Siemens) for safety-critical environments, leading end-to-end development from architecture and electrical design to on-site commissioning.
- Engineered control panels, HMI–PLC solutions, and sensor integration pipelines for compressors, hydraulic machinery, and power conditioning systems.
- Implemented safety logic, fault detection, and operational interlocks for high-power electrical and hydraulic systems operating in 24/7 industrial plants.

3D-Printed Upper-Limb Prosthesis (2024).
Low-cost prosthesis designed and fabricated for a patient as part of a community service initiative.

REFERENCES

Dr. Christian Tutiven Galvez — Head of CoRAL, ESPOL, Ecuador
cjtutiven@espol.edu.ec

Prof. David Garzón Ramos — Assistant Professor, UCD, Ireland
david.garzonramos@ucd.ie

Prof. Nabih Pico — Research Professor, RISE Lab, SKKU, South Korea
npico@g.skku.edu

PUBLICATIONS & INTELLECTUAL PROPERTY

Modular Robotic Platform and Manipulation System for Swarm Construction Robots 2025
Industrial secret (filing stage). Co-inventor of a modular robotic actuator for swarm platforms capable of collective manipulation and construction. Contributions: mechanical and electromechanical actuator design, PCB schematics, CAD modeling, and integration with the mobile robotic platform.

LEADERSHIP & SERVICE

Founder Member and Former Vice President — Artificial Intelligence and Robotics Society club (AIRoS) 2024 – 2025
Escuela Superior Politécnica del Litoral (ESPOL), Ecuador

- Co-founded AIRoS, a student-led robotics and AI research group. Recruited and mentored undergraduate students in ROS-based robotic systems and autonomous navigation.
- Led technical training, project-based learning initiatives, and coordinated academic outreach events featuring international PhD researchers.
- Led the team that won first place at Roboracer Ecuador (autonomous driving competition); a second AIRoS team placed second.
- Helped establish a research culture that produced 3 conference papers and 2 journal articles by student members in one year.

EDUCATION

B.Sc. in Mechatronics Engineering 2025
Escuela Superior Politécnica del Litoral (ESPOL), Ecuador

- Thesis: *Swarm Robotic Manipulation and Collective Transport for Construction-Oriented Tasks*

English Proficiency
Test of English as a Foreign Language (TOEFL iBT) 2025
Score: 86